DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

SECP3213-02 - BUSINESS INTELLIGENCE

--

PHASE 2 : Data Workflow (Alteryx Designer)

--

TITLE : Retail Sales Performance and Customer Behaviour Analysis

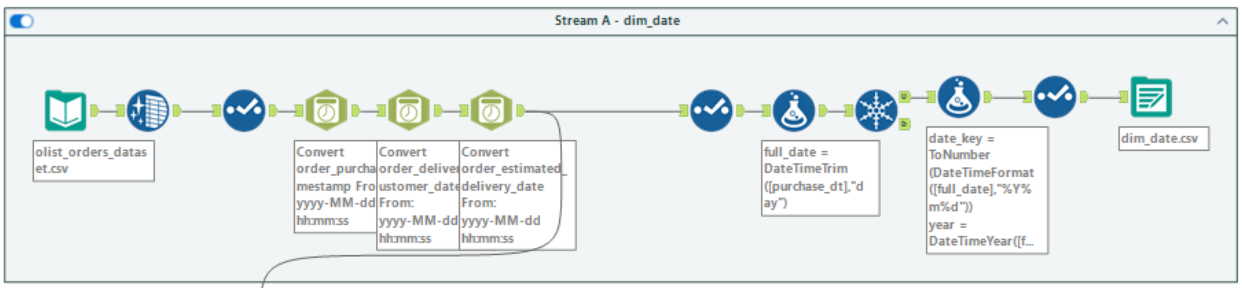
PREPARED BY : ALLSHAR

NAME	**MATRIC NO**
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134
IMAN ABADI BIN MOHD NIZWAN	A23CS0084
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112

Workflow Methodology

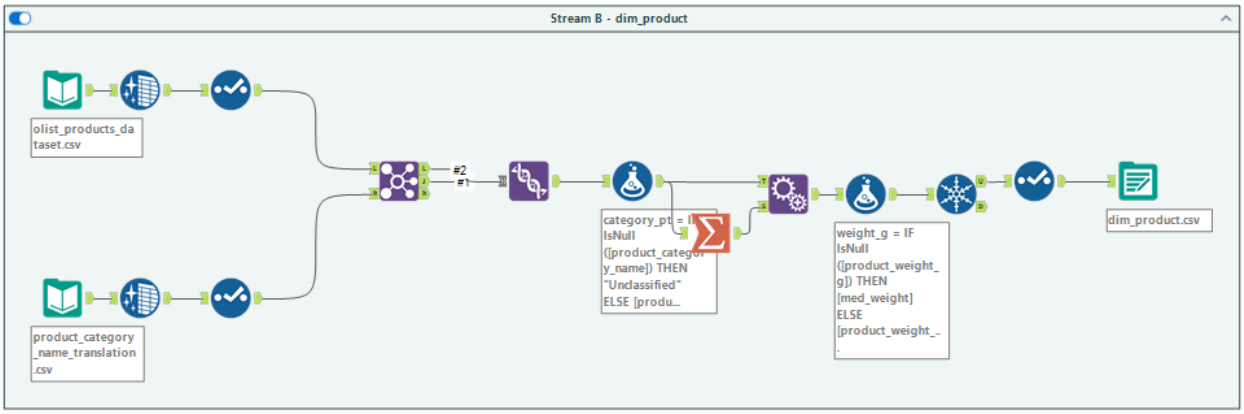
1. Stream A : Date Dimension (dim_date)

Process	First, the purchase times are turned into a standard date format and shortened so they only show the day. Then, duplicate dates are removed so there is only one row for each day. After that, a formula tool creates a readable date code like (YYYYMMDD), along with separate columns for the year, month, month name, and quarter.
Tools	Input Data, Select, DateTime, Formula, Unique, Output Data.
Result	A date dimension of ~634 rows covering Sep 2016 – Oct 2018.



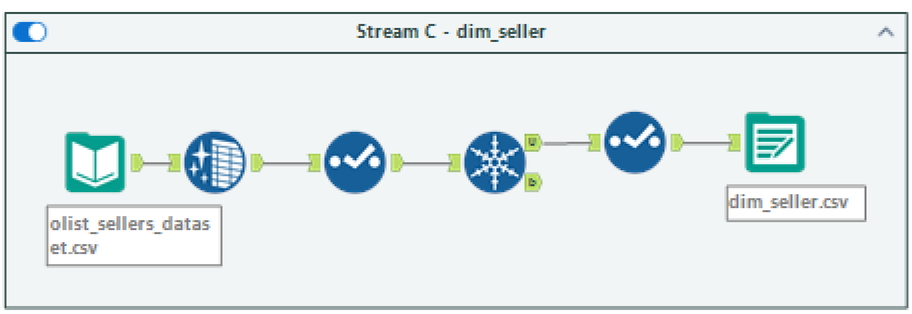
2. Stream B : Product Dimension (dim_product)

Process	The products table is connected to the category translation table using the product category name. Both matched and unmatched rows are put back together so no products are dropped. For the products with missing category names, the blank spaces are filled in with the word "Unclassified." If a product is missing its weight, it gets filled in with the middle weight value of the whole column. Finally, a new column is made to calculate the total size of each product by multiplying its dimensions together.
Tools	Input Data, Join, Union, Formula, Summarize, Append Fields, Unique, Select, Output Data.
Result	~32,951 products with complete categories and weights.



3. Stream C : Seller Dimension (dim_seller)

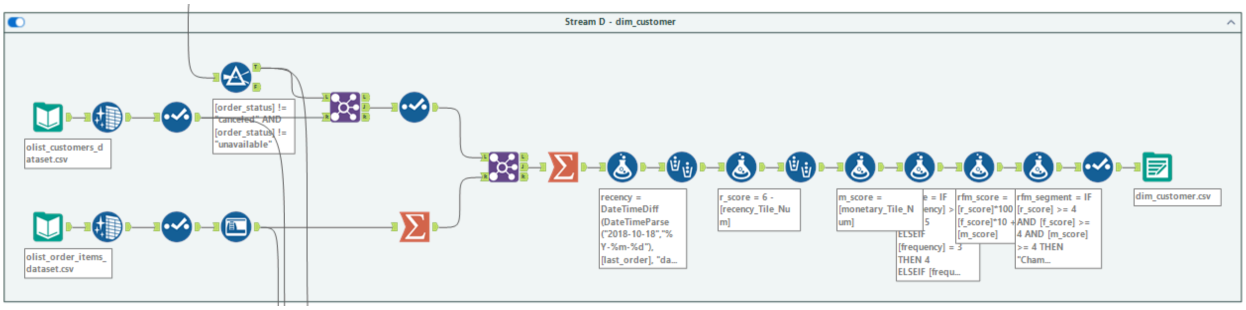
Process	Seller city and state are standardized with the Data Cleansing tool (whitespace removal). The Unique tool guarantees one row per seller_id.
Tools	Input Data, Data Cleansing, Unique, Select, Output Data.
Result	~3,095 unique sellers.



4. Stream D : Customer Dimension with RFM (dim_customer)

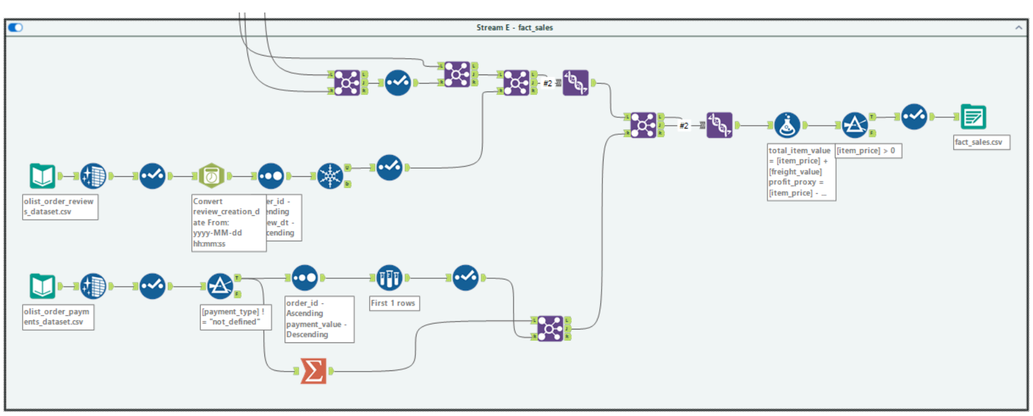
Process	Valid orders are joined to the customers table to obtain customer_unique_id and to perorder revenue aggregated from order items. A Summarize tool aggregates the data to one row per customer, computing last order date, distinct order count (Frequency) and total spend (Monetary). Recency is the days between the last order and a fixed snapshot of 18 October 2018. The Multi-Field Binning tool assigns quantile scores for Recency and Monetary. Frequency uses manual bins. A Formula tool produces the composite rfm_score and a named rfm_segment.
Tools	Input Data, DateTime, Filter, Join, Summarize, Formula, Multi-Field Binning, Select, Output Data.

Result	~94,990 customers scored and segmented across all five groups.



5. Stream E : Sales Fact Table (fact_sales)

Process	Reviews are deduplicated (sort by date descending, then Unique on order_id) to keep the latest of the 551 duplicated orders. Payments are aggregated to one total value per order, with the dominant payment type selected using Sort-and-Sample. Order items (with price renamed to item_price) are joined to valid orders, then to the customer identifier, the deduplicated reviews, and the aggregated payments. The left-unmatched rows are kept through Union so items without a review or payment are not dropped. Formula tools derive total_item_value, profit_proxy (price minus freight, a contribution proxy), delivery_days (null-guarded), and order_date_key.
Tools	Input Data, DateTime, Sort, Unique, Filter, Sample, Summarize, Select, Join, Union, Formula, Output Data.
Result	~112,101 order-item rows. All foreign keys resolve to a dimension row.



Tools Used

Tool	Purpose
Input / Output Data	Read 9 CSVs, write 5 tables
Select	Trim columns, rename, set types
DateTime	Parse text to DateTime
Filter	Remove canceled/unavailable, zero price
Formula	Derived fields and RFM scores
Summarize	Aggregate to order/customer grain
Tile	Quantile scoring (R, M)
Join	Combine datasets on keys
Union	Preserve unmatched rows
Unique	Deduplicate reviews/sellers/dates
Sort + Sample	Dominant payment type per order
Append Fields	Broadcast median to all rows
Data Cleansing	Broadcast median to all rows

Challenges

Working with the Olist dataset was challenging because of a few hidden problems missed and failed to be identified in the proposal. First, mixing up a customer's temporary order ID with their real, permanent ID would completely ruin the customer groups. Second, almost everyone only bought something once, only about 2,888 out of nearly 95,000 customers came back a second time. Because of this, standard automated scoring for how often people shop did not work, so the groups had to be defined manually. Also, different tables tracked data differently, like overall orders versus single items, so order details had to be combined first to avoid counting things twice. Another issue was that there were orders that had more than one review, so the data had to be sorted to keep only the newest one. Finally, because the dataset does not show the actual cost of the products, profit was computed by just subtracting shipping fees from the item price. Since this is just a rough guess and not true profit, any financial insights should be looked at carefully.